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# Reasoning Backwards from Evidence

*Bayes' Theorem, Updating Beliefs & the Naïve Bayes Classifier*

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**Prepared by Prof. Saurabh**

We usually know things forwards: *given the cause*, how likely is the evidence? But life asks the reverse: *given the evidence*, how likely is the cause? Bayes' theorem flips the question around. This companion builds it from the rules of Session 3, shows how a prior belief gets *updated* into a posterior, and then turns it into one of machine learning's most useful tools — the Naïve Bayes classifier that filters spam, tags topics, and predicts who will play tennis.

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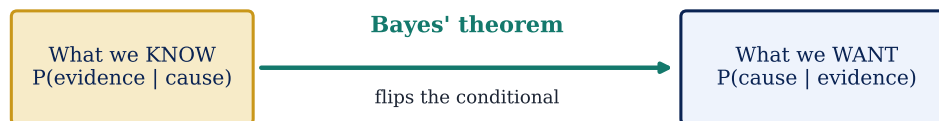
## 1 The question runs backwards

Session 3 took us *forwards*: given a cause, how likely is some evidence?  $P(\text{evidence} \mid \text{cause})$  is usually easy to measure — “80% of spam emails contain the word *offer*.” But the question we actually care about runs the other way: an email *contains* “offer” — how likely is it spam? That is  $P(\text{cause} \mid \text{evidence})$ , and Bayes’ theorem is the tool that flips one into the other.

### What, why, where — for this whole session

**What:** a formula that *reverses* a conditional probability, turning “evidence given cause” into “cause given evidence,” and a classifier built on it. **Why:** we can only *observe* effects (symptoms, words, clicks) but want to know the hidden *cause* (disease, spam, intent) — exactly the direction Bayes provides. **Where:** spam filters, medical diagnosis, sentiment analysis, document tagging, and as the beating heart of the Naïve Bayes classifier used across machine learning.

Bayes' theorem turns the question around



*Bayes' theorem reverses the arrow: it converts the thing we can easily measure,  $P(\text{evidence} \mid \text{cause})$ , into the thing we want,  $P(\text{cause} \mid \text{evidence})$ .*

### The doctor and the test

A test detects a disease 99% of the time when it is present — that is  $P(\text{positive} \mid \text{disease})$ , measured in the lab. But a patient who just tested positive wants  $P(\text{disease} \mid \text{positive})$ . These two are *not* the same number, and confusing them is a famous, dangerous mistake. Bayes’ theorem keeps them straight.

## 2 Bayes’ theorem, built from what we already know

Recall the multiplication rule from Session 3, written two ways for the same overlap:

$$P(A \cap B) = P(A) P(B \mid A) = P(B) P(A \mid B).$$

Set the two right-hand sides equal and divide by  $P(B)$ . That is the whole derivation.

**Key formula 2.1: Bayes' theorem**

$$P(H | E) = \frac{P(E | H) P(H)}{P(E)}.$$

$P(H)$  — **prior**: what we believed about the hypothesis  $H$  *before* the evidence.

$P(E | H)$  — **likelihood**: how well  $H$  predicts the evidence  $E$ .

$P(H | E)$  — **posterior**: our *updated* belief after seeing  $E$ .

$P(E)$  — **evidence**: a normaliser, found by total probability over all hypotheses:  $P(E) = \sum_i P(E | H_i) P(H_i)$ .

**A belief-updating machine**

Read Bayes as: *new belief*  $\propto$  *old belief*  $\times$  *how well it explains the evidence*. Strong prior belief can be overturned by strong evidence, and weak evidence barely nudges a strong prior. It is the mathematics of changing your mind by exactly the right amount.

**Worked example 2.1: A first Bayes calculation: rash, flu or measles?**

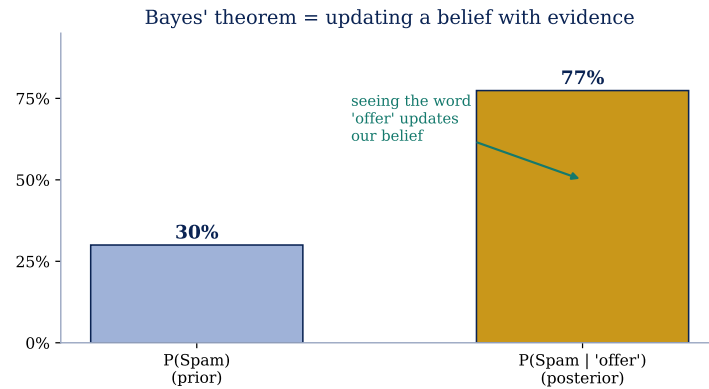
In a neighbourhood, 90% of sick children have flu and 10% have measles. A rash appears in 95% of measles cases but only 8% of flu cases. A child develops a rash — is it flu? First the evidence probability (total probability over the two diseases):

$$P(\text{rash}) = \underbrace{(0.08)(0.90)}_{\text{flu path}} + \underbrace{(0.95)(0.10)}_{\text{measles path}} = 0.072 + 0.095 = 0.167.$$

Then flip with Bayes:

$$P(\text{flu} | \text{rash}) = \frac{(0.08)(0.90)}{0.167} = \frac{0.072}{0.167} \approx 0.43, \quad P(\text{measles} | \text{rash}) \approx 0.57.$$

Striking: although flu is nine times more common, the rash is such strong evidence for measles that, once we see it, *measles becomes the more likely cause*. That is the prior and the likelihood fighting — and the likelihood winning.



Before seeing any words, an email is spam with probability 0.30 (the prior). After we notice the word “offer”, Bayes lifts that belief to 0.77 (the posterior). Evidence moved the needle.

### Worked example 2.2: Spam filter, one word

30% of emails are spam. The word “offer” appears in 80% of spam but only 10% of normal email. An email contains “offer” — is it spam? First the evidence probability (total probability):

$$P(\text{offer}) = \underbrace{(0.8)(0.3)}_{\text{spam path}} + \underbrace{(0.1)(0.7)}_{\text{not-spam path}} = 0.24 + 0.07 = 0.31.$$

Then Bayes:

$$P(\text{spam} | \text{offer}) = \frac{(0.8)(0.3)}{0.31} = \frac{0.24}{0.31} \approx 0.77.$$

So “offer” raises the spam probability from 30% to about 77% — a big, but not absolute, update.

### Don't forget the base rate (the most common Bayes mistake)

A test being “99% accurate” does *not* mean a positive result is 99% likely to be real. If the thing you're testing for is *rare* (a low prior, or *base rate*), then the few false alarms from the huge healthy majority can easily outnumber the true detections — so a positive result may still be *probably* wrong. Bayes' theorem handles this automatically through the prior  $P(H)$ ; our intuition usually doesn't. This is exactly why practice problem 2 below feels surprising.

**Worked example 2.3: The base-rate surprise, worked out**

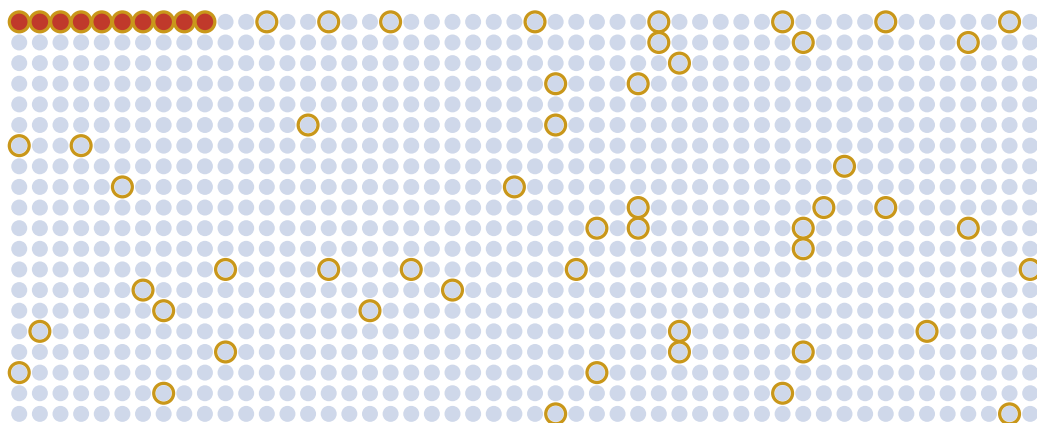
A disease affects 1% of people. A test catches 99% of true cases ( $P(+ | D) = 0.99$ ) but also wrongly flags 5% of healthy people ( $P(+ | D^c) = 0.05$ ). You test positive — how worried should you be? Total probability for the evidence:

$$P(+) = (0.99)(0.01) + (0.05)(0.99) = 0.0099 + 0.0495 = 0.0594.$$

$$P(D | +) = \frac{(0.99)(0.01)}{0.0594} \approx 0.17.$$

Only about 17%! Because the disease is rare, the flood of false positives from the healthy 99% swamps the few true positives. A “99% accurate” test on a rare condition is still *probably wrong* when it fires — a result that saves real lives once doctors internalise it.

Why a 99%-accurate test can still be usually wrong: imagine 1000 people

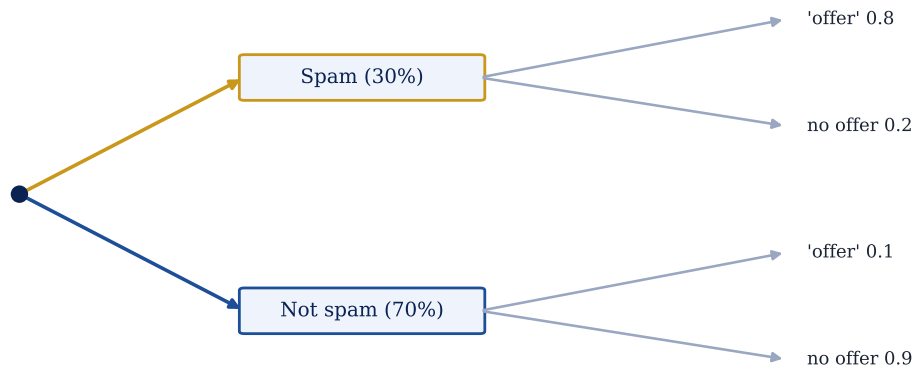


Of ~60 positives, only 10 are truly sick →  $P(\text{disease} | \text{positive}) \approx 10/60 \approx 17\%$

● has disease (10)    ● healthy (990)    ○ tests positive (gold ring)

The same result without a single formula — just count people. Among 1000, about 10 are truly sick (red) and almost all test positive; but 5% of the 990 healthy also test positive ( $\approx 50$  gold rings). So of  $\approx 60$  positives only 10 are real — about 17%. Switching from probabilities to natural frequencies makes Bayes feel obvious.

## Reading the spam filter as a probability tree



Each path multiplies its probabilities; Bayes flips it to get  $P(\text{Spam} \mid \text{'offer'})$

*The same calculation as a tree: split on spam/not-spam (the prior), branch on the word (the likelihood), multiply along each path, and let Bayes flip the tree to answer the backwards question.*

### 3 From one feature to many: the Naïve Bayes classifier

Real classifiers use *many* clues at once — dozens of words, several symptoms, multiple features  $X_1, \dots, X_n$ . Bayes still applies, but  $P(X_1, \dots, X_n \mid C)$  is impossible to estimate honestly: there are far too many combinations. So we make one bold, simplifying assumption.

#### Definition 3.1: Conditional independence — the “naïve” assumption

Naïve Bayes assumes the features are **independent once the class is known**:

$$P(X_1, \dots, X_n \mid C) = \prod_{i=1}^n P(X_i \mid C).$$

That is, within a class, each feature contributes its own little vote and the votes don’t interfere.

#### Why “naïve”?

The word “offer” and a suspicious link probably *do* tend to occur together in spam — they are not really independent. Naïve Bayes cheerfully ignores that and treats every clue as if it arrived on its own. It is “naïve” for assuming this, yet astonishingly effective in practice — like a committee that votes independently and still usually reaches the right verdict.

**Key formula 3.1: The Naïve Bayes decision rule (MAP)**

To classify a new point with features  $X_1, \dots, X_n$ , pick the class  $y_k$  that maximises

$$P(Y = y_k) \prod_i P(X_i | Y = y_k).$$

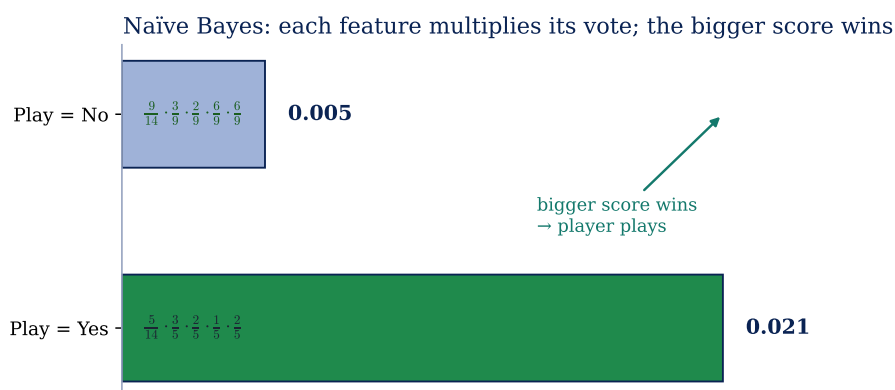
The shared denominator  $P(X_1, \dots, X_n)$  is the same for every class, so we can ignore it and just compare the numerators — “most probable class wins.”

**Worked example 3.1: Will the player play tennis?**

From 14 past days we tally how often each weather feature occurred on Play = Yes vs. No days, with  $P(\text{Yes}) = \frac{9}{14}$ ,  $P(\text{No}) = \frac{5}{14}$ . For Today = (Sunny, Hot, Normal, not Windy) we multiply each feature's conditional probability by the class prior:

$$\text{Yes-score} = \frac{3}{9} \cdot \frac{2}{9} \cdot \frac{6}{9} \cdot \frac{6}{9} \cdot \frac{9}{14} \approx 0.021, \quad \text{No-score} = \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{2}{5} \cdot \frac{5}{14} \approx 0.005.$$

Since the Yes-score is larger, we predict **Yes — the player will play**. We never needed the exact probabilities, only which class scored higher.



*Naïve Bayes as a vote-off. Each class starts with its prior and then multiplies in one factor per feature. The class with the larger product wins — here “Yes” (0.021) comfortably beats “No” (0.005). The denominator  $P(X)$  is identical for both classes, so it never affects who wins and we simply drop it.*

**The zero-frequency trap and Laplace smoothing**

What if a word never appeared in a class in the training data? Its conditional probability is 0, and because we *multiply* the features, one zero wipes out the entire score — no matter how strong the other clues are. The fix is **Laplace (add-one) smoothing**: add 1 to every count so no probability is ever exactly zero. It is the gentle admission that “we just haven’t seen it *yet*” is different from “it is impossible.”



**Worked example 3.2: “Lunch Money Money Money” — why smoothing is needed**

A spam tally has the word “Money” appearing 4 times in spam but 0 times in normal mail. Without smoothing,  $P(\text{Money} \mid \text{normal}) = 0$ , so *any* message containing “Money” would be forced to a normal-score of 0 — clearly wrong for a money-spam blast. Adding 1 to each count gives every word a small, non-zero chance, and the classifier then correctly flags the message as spam.

**Your turn**

1. An academician, businessman, and politician become Vice-Chancellor with probabilities 0.5, 0.3, 0.2, and promote research with probabilities 0.8, 0.6, 0.4. Research *was* promoted — what is the probability the VC was the academician? (*Total probability for the denominator, then flip with Bayes.*)
2. A spam detector flags 99% of spam and has a 5% false-positive rate. If half of all email is spam, and a message is flagged, what is the probability it is actually *not* spam? (*The answer surprises most people — that’s the point.*)
3. Using a tiny labelled corpus, classify the sentence “A very close game” as Sports or Not-Sports with Naïve Bayes, applying Laplace smoothing so the unseen word “close” doesn’t zero out the Sports score.

**The thread to the next session**

Bayes and Naïve Bayes treat each feature as a value with a probability attached. To handle features that are numbers — heights, waiting times, counts — we need to describe *how a numerical quantity spreads its probability*. That object is a **random variable** and its **probability distribution**, the subject of Session 5.